# Emil I. Jaffal

ejaffal@gradcenter.cuny.edu | emiljaffal | emiljaffal.github.io | emiljaffal | emiljaffal

## Education

**City University of New York**, The Graduate Center — Expected 2028
*Ph.D., Chemistry*
Advisor: Dr. Anton Oliynyk

**Fordham University**, Fordham College at Rose Hill — May 2023
*B.Sc., Chemistry*

## Research Experience

**Ph.D. Student** — *New York, NY*
*City University of New York, Hunter College* — Jul 2024 – Present

**Solid-State Laboratory – Dr. Anton Oliynyk**

- Conducting exploratory syntheses of novel intermetallic materials with corresponding analyses using powder X-ray diffraction and scanning electron microscopy.
- Enhancing machine learning applications to predict properties of various binary and ternary compounds, focusing on improving interpretability and predictive capabilities of models in solid-state materials by incorporating detailed structural information.
- Directly mentoring a handful of students, providing guidance in their research projects with both experimental and computational techniques.

**Undergraduate Researcher** — *Bronx, NY*
*Fordham University* — Sep 2021 – May 2023

**Organic/Materials Laboratory – Dr. Julia Schneider**

- Steered materials research involving various reactions as part of a novel multi-step synthesis to create organic semiconductors (OSCs) with tunable conjugated heterocycles to improve conductivity.
- Instrumentation experience includes handling UV-Vis, NMR, fluorescence, and IR spectroscopy with respective machinery and analytic interpretations. General synthesis and purification skills include distillations, extractions, filtrations, and recrystallizations.

**Computational Laboratory – Dr. Joshua Schrier**

- Identified probable transition states of novel syntheses as part of an NSF-funded collaboration within the chemistry department.
- Performed numerous Gaussian ab initio calculations of internal energies, electronic structures, and geometric data using density functional theory to analyze reaction thermodynamics and predict isomer formations of OSCs.

## Professional Experience

**Research Chemist** — *Tarrytown, NY*
*ICL Industrial Products* — Sep 2023 – Jul 2024

- Developed novel flame retardant (FR) blends for polyurethane foams in collaboration with external manufacturers and customers, ensuring compliance with international safety regulations.
- Pioneered the integration of polyurethane for battery encapsulation, contributing to advancements in sustainable and high-performance materials, leading to a patent application.
- Led application efforts for VeriQuel® F series, a proprietary phosphorus-based FR for flexible foams,

conducting iterative testing with customers and scaling toll production for MT quantities, with early sales reaching $800K.

- Executed laboratory experiments, standardized flammability tests, and physical property assessments to support new product development and market-driven innovations in halogenated and non-halogenated FRs.

## Publications

**Indicates corresponding author*

**15. Achieving a Scalable Machine Learning Workflow for Crystal Structure Discovery with Experimental Validations**. *In preparation.* Mar 2026
Shiryaev D., **Jaffal E.I.**, Lee S., Selvaratnam B., & Oliynyk A.O.

**14. Systematic Ternary Exploration and Expansion of the Ternary Dy–Ir–Sn System with Magnetic Investigation**. *In preparation.* Mar 2026
**Jaffal E.I.**, Brodska A., Selvaratnam B., & Oliynyk A.O.

**13. Systematic Ternary Exploration and Expansion of the Ternary Gd–Os–Ge System**. *In preparation.* Mar 2026
**Jaffal E.I.**, Oziel J., Selvaratnam B., & Oliynyk A.O.

**12. Evaluation of Machine Learning Models in the Search of Interpretable Structure Maps of Perovskites and Spinels through Modified SISSO Routine**. *Preprint submitted, Solid State Commun.* Mar 2026
Selvaratnam B., Oziel J., **Jaffal E.I.**, & Oliynyk A.O.

**11. Disentangling structural patterns in quaternary germanides with materials informatics and exploratory synthesis: $RE_4$IrInGe$_4$ (RE = Y, Ce–Nd, Sm, Gd, Ho–Er) series**. *Accepted, Inorg. Chem.* Mar 2026
Ponomarev A., Selvaratnam B., **Jaffal E.I.**, Shiryaev D., & Oliynyk A.O.

**10. Improving Thermal Cycling Stability and Magnetocaloric Efficiency through Orbital Competition and Entropy Engineering in $RE_{11}$Rh$_4$In$_9$, 4*d*-4*f* Layered Intermetallics**. *Chem. Mater.* Feb 2026
Tyvanchuk Y., ..., **Jaffal E.I.**, Selvaratnam B., Mudring A., & Oliynyk A.O.

**9. Exploring Feature Engineering for Crystal Structure Classification: Interactive Applications of PCA & PLS-DA**. *J. Chem. Educ.* Jan 2026
Shiryaev D., **Jaffal E.I.**, Selvaratnam B., Sun Y. & Oliynyk A.O.

**8. $RE_7$M$_4$InGe$_{12}$ ($RE$ = Y, Gd–Er; $M$ = Rh, Os) Quaternary Germanide Series Expansion with Structural Motif Clustering**. *Inorg. Chem.* Dec 2025
Pozdnyakova N., **Jaffal E.I.**, ... & Oliynyk A.O.

**7. Investigating Mechanical Properties Through Defect Chemistry in Hard Binary Phosphide Material Ta$_3$P.** *Solid State Commun.* Nov 2025
**Jaffal E.I.**,* Shiryaev D., Selvaratnam B. & Oliynyk A.O.

**6. Dataset of Prototype Structures Adopted by Intermetallic Compounds with AB Stacking**. *Data Brief.* Oct 2025
Selvaratnam B., **Jaffal E.I.**, Shiryaev D. & Oliynyk A.O.

**5. Explainable Recommendation Engines to Predict Complex Intermetallics: Synthesis & Characterization of Gd$_{10}$RuCd$_3$, a Neutron Absorption Material.** *J. Am. Chem. Soc.* Sep 2025
Xhabrahimi B., **Jaffal E.I.**, ... & Oliynyk A.O.

**4. Materials Informatics Tools to Analyze Crystal Structures: Crystal Structure of Novel Ternary Indide $ErCo_2In$**. *Integr. Mater. Manuf. Innov.* Jun 2025
Tyvanchuk Y., Lee S., ..., **Jaffal E.I.**, Selvaratnam B. & Oliynyk A.O.

**3. Unsupervised ML Prediction of Novel 1:3 Intermetallic with Synthesis of $TbIr_3$ ($PuNi_3$-type) as Experimental Validation**. *J. Am. Chem. Soc.* Feb 2025
Sethi S.S., Dutta A., **Jaffal E.I.**, ... & Oliynyk A.O.

**2. Composition and Structure Analyzer/Featurizer for Explainable ML Models to Predict Solid State Structures**. *Digit. Discov.* Jan 2025
**Jaffal E.I.**, ... & Oliynyk A.O.

**1. Synthesis of PyrDi Isomers w/ Tunable Excimer Formation**. *Org. Lett.* Jan 2025
Johnston K., McCostis A., Mikita E., **Jaffal E.** & Schneider J.A.

## Guest Lectures

Solid-State Chemistry: *Introduction to Thermoelectrics – Hunter College, NY* May 2025

## Courses Taught

General Chemistry Lab (10600) – *Hunter College, NY* Aug 2025 – Present

## Presentations

**Brookhaven Lab Nuclear Chemistry Summer School** – *New York, NY* Jul 2025
Using explainable recommendation engines for the discovery of $Gd_{10}RuCd_3$

**ACS Mid-Atlantic Regional Meeting** – *South Orange, NJ* May 2025
Quaternary Intermetallic Germanides: Structure, Properties, and Potential Applications

**Fordham University Jean Dreyfus Lectureship** – *Bronx, NY* Apr 2023
The Schneider Lab

**Brookhaven Lab Nuclear Chemistry Summer School** – *New York, NY* Jul 2024
The Oliynyk Lab

**MAPS: Research at Fordham** – *Bronx, NY* Nov 2022
Vinyl Azide Cyclization: Where Organic and Computational Chemistry Meet

## Posters

**North American Solid State Chemistry Conference** – *Ames, IA* Jul 2025
Quaternary Intermetallic Germanides: Structure, Properties, and Potential Applications

## Hackathons

**SSMC-Collaboration Incubator** – *Madison, WI* May 2025

Selected participant for national hackathon-style research workshop. Collaborated on the *Rational Design of Thermoelectrics* with an interdisciplinary cohort of PhD students and professors.

## Selected Projects

*See the full suite of apps on the Oliynyk lab website, both developed myself here.*

**XRD Comparison & Matching Tools** Mar 2025

Contributed to the development and deployment of GUIs, allowing novice users to compare multiple '.xy' files from X-ray diffraction (XRD) data with .cif files. These help identify impurities, match phases, and visualize differences in crystal structure XRD patterns with ease.

**Composition Analyzer/Featurizer (CAF)** Jun 2024

Developed an interactive Python script that generates chemical compositional features and provides tools for filtering, sorting, and merging data. Aids novice solid-state chemists and materials scientists in generating compositional training data for machine learning models ranging from dozens to tens of thousands of compounds.

## Patents

**Gadolinium–ruthenium–cadmium intermetallic compound with negative thermal expansion and neutron-absorbing uses thereof**. No. 63/997,273. — Mar 2026

**Emil Jaffal**, Anton Oliynyk & Balaranjan Selvaratnam. *Patent pending.*

**Heat Resistant Semi-Rigid Polyurethane Foams**. *WO 2026/035457 A1*. — Feb 2026

**Emil Jaffal**, Sergei Levchik, Zhihao Chen, Jeffrey Stowell & Munjal Patel.

## Awards and Grants

CUNY Provost's Pre-Dissertation Science Research Fellowship — 2026
*Awarded $4,000 to support early-stage summer research for the puprposes of dissertation proposal development.*

CUNY Doctoral Student Research Grant (DSRG) — 2026
*Awarded $1,239 to support travel for training in uranium intermetallic synthesis, as part of dissertation research and professional development.*

CUNY Black and Schwartz Science Conference Travel Award — 2026
*Awarded a prestigious $1,400 grant to attend a conference to present research.*

NY-Hub Regional I-Corps Virtual Innovation Lean Bootcamp — 2026
*Selected for the Winter 2026 cohort with access to a $3,000 customer discovery reimbursement fund for entrepreneurial development and startup experience.*

Hunter College Cliff Sol Award — 2025
*Awarded $500 for excellence by a graduate student.*

CUNY Graduate Center Certificate of Achievement — 2025
*Awarded $1,000 in recognition of outstanding success as a first-year student, including various publications during my first year.*

CUNY Graduate Center Certificate of Achievement — 2025
*Awarded $3,000 in recognition of the acceptance of a first-author publication during the Fall 2024 semester.*

CUNY Graduate Center Science Scholarship — 2024

Fordham University Dean's List — 2023

NSF Summer Research Funding Grant (DMR-1928882) — 2022

## Service

*Content Creator of Educational Content* — Jun 2025 – Present
Founder/manager of the Oliynyk Lab YouTube channel for educational outreach.

Materials Today Physics – *Reviewer* — Aug 2025 – Present

Fordham University Muslim Students Association – *Treasurer* — Sep 2022 – May 2023

Fordham University Arabic Club – *Vice President* — Jan 2022 – Aug 2022

Fordham Undergraduate Research Journal – *Peer Editor* — Sep 2022 – May 2023

## Students Mentored

**Raoul Guillet** (B.Sc. Chemistry, 2026)
**Alexandra Brodska** (B.Sc. Biology, 2027)
**Brook Xhabrahimi** (B.Sc. Chemistry, 2025)
**Natalia Poznyakova** (B.Sc. Chemistry, 2025)
**Miriam Ismail** (B.Sc. Chemistry, 2025)
**Riya Upadhyay** (B.A. Human Biology, 2024)
**Alex Vtorov** (B.Sc. Chemistry, 2025)
**Joseph Oziel** (The Bronx High School of Science, 2025)
**Yujing Sun** (The Bronx High School of Science, 2025)
**Brandon Lin** (The Bronx High School of Science, 2025)

## Memberships

Sigma Xi, The Scientific Research Honor Society – *Associate Member* Mar 2023 – Present

## Technical skills

**Software:** Adobe Illustrator, Bluehill, ChemOffice, Gaussian16, Mathematica, Maestro, Microsoft Suite, Signals Notebook, TopSpin, VASP, WebMO.
**Programming & markup languages:** Python, Bash, HTML, Wolfram (Mathematica).
**Packages:** NumPy, SciPy, Scikit-Learn, Pandas, Matplotlib.
**Languages:** English (native), Arabic (native), Spanish (conversational).